Principles of Discrete Applied Mathematics

Personal Notes

Mihir Rao

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1 The Pigeonhole Principle

The pigeonhole principle is among the simplest statements in mathematics and among the most surprisingly powerful. It asserts nothing more than that objects cannot be distributed among fewer containers without doubling up somewhere, yet it settles questions that look nothing like counting problems. This lecture states the principle and then applies it to graphs, to subset sums, to monotone subsequences, and to the information-theoretic limits of a guessing game.

Begin with a puzzle. A mathematician gets up in the dark and, to avoid waking their partner, dresses in another room. They grab socks from a drawer that contains socks of only two colors. How many must they take to guarantee a matching pair?

The answer is three. Two socks may well be mismatched, but with three socks and only two colors some color must appear twice. The general shape of the argument is already visible: when there are more socks than colors, two socks share a color.

1.1 Statement

Definition: If m objects (the *pigeons*) are placed into n boxes (the *pigeonholes*) and $m > n$, then at least one box contains more than one object.

A stronger form gives a quantitative guarantee rather than a bare existence claim.

Theorem 1.1 (Pigeonhole principle, strong form)**.** *If m objects are placed into n boxes, then at least one box contains at least $\lceil \frac{m}{n} \rceil$ objects.*

Here $\lceil \cdot \rceil$ denotes rounding up to the next integer, and $\lfloor \cdot \rfloor$, used later, denotes rounding down. The basic form is the case $m > n$, where $\lceil m/n \rceil \geq 2$.

> ***Note:*** *The principle is so elementary that it is natural to suspect nothing interesting can follow from it. That suspicion is wrong. The difficulty in applying it is never the principle itself; it is choosing what to call a pigeon and what to call a pigeonhole.*

1.2 Two Vertices of Equal Degree

A *graph* G consists of a set V, whose elements are called *vertices*, and a set E of pairs of vertices, whose elements are called *edges*. The *degree* of a vertex v is the number of edges containing v.

Theorem 1.2. *In any finite graph, there are two vertices of equal degree.*

Told as a story: at a party with n people, there are two people who know the same number of people at the party.

Proof. Let the graph have n vertices. Every degree is an integer between 0 and $n-1$, which is n possible values for n vertices — so the counting alone does not yet force a repeat. Suppose for contradiction that all degrees are distinct. Then each of the n possible values occurs exactly once. In particular some vertex v has degree 0, and some vertex w has degree $n-1$, meaning w is adjacent to every other vertex.

These two cannot coexist. If the edge (v, w) is present, then v has a neighbor and its degree is not 0. If the edge (v, w) is absent, then w fails to be adjacent to v and its degree is not $n-1$. Either way we have a contradiction, so two degrees must coincide. □

Note: *The pigeons are the n vertices and the pigeonholes are the n possible degrees, so the counts match exactly and the principle does not apply directly. The work of the proof is to rule out one of the n holes — degrees 0 and $n-1$ cannot both be occupied — which leaves n pigeons in $n-1$ usable holes.*

1.3 Equal Sum Subsets

Here the principle does something less obvious.

Theorem 1.3. *Among any 30 seven-digit numbers there exist two disjoint nonempty subsets with the same sum.*

Proof. Take the nonempty subsets as pigeons: there are $2^{30}-1$ of them. Take the possible subset sums as pigeonholes. Each number is less than 10^7, so the sum of at most thirty of them is at most $3 \cdot 10^8$, which is smaller than $2^{30}-1 \approx 10^9$. There are more subsets than achievable sums, so two distinct subsets A and B satisfy

$$\sum_{x \in A} x = \sum_{x \in B} x.$$

These subsets need not be disjoint, but the common elements can be discarded. Put $A' = A \setminus B$ and $B' = B \setminus A$. Removing the same quantity $\sum_{x \in A \cap B} x$ from both sides preserves the equality of sums, so A' and B' have equal sums and are disjoint by construction. Since A and B were distinct, at least one of A', B' is nonempty — and equal sums then force the other to be nonempty as well. □

Note: *The proof shows the two subsets* exist *but gives no way to find them. For 30 seven-digit numbers a search is within reach of modern computers, but at 100 numbers of correspondingly greater size, every known method takes an enormous amount of time. Existence and construction are genuinely different problems, and pigeonhole arguments deliver only the former.*

1.4 Monotone Subsequences

A *subsequence* of $a_1, a_2, \ldots, a_k$ is what remains after deleting some terms and keeping the rest in their original order; it is *monotone* if it is entirely increasing or entirely decreasing. Long sequences of distinct reals are forced to contain long monotone subsequences.

Consider the sequence of length 12

$$4,\ 3,\ 2,\ 1,\ 8,\ 7,\ 6,\ 5,\ 12,\ 11,\ 10,\ 9.$$

The terms $4, 8, 11$ form an increasing subsequence of length 3, and $4, 3, 2, 1$ form a decreasing subsequence of length 4. Neither can be improved within this sequence. The next theorem explains why it stops there: with $m = 3$ and $n = 4$ the sequence has exactly $mn = 12$ terms, one short of the $mn + 1 = 13$ that would force an increasing subsequence of length 4 or a decreasing subsequence of length 5.

Theorem 1.4. *Any sequence of $mn+1$ distinct real numbers $a_1, a_2, \ldots, a_{mn+1}$ has an increasing subsequence of length $m + 1$ or a decreasing subsequence of length $n + 1$.*

Proof. Let $t(i)$ denote the length of the longest increasing subsequence ending at a_i. If $t(i) > m$ for some i, an increasing subsequence of length at least $m + 1$ exists and we are done. So assume $t(i) \in \{1, 2, \ldots, m\}$ for every i.

Now apply the principle: the $mn + 1$ indices are the pigeons and the m possible values of t are the pigeonholes. Since $\lceil (mn + 1)/m \rceil = n + 1$, some value s satisfies $t(i_j) = s$ for at least $n + 1$ indices $i_1 < i_2 < \cdots < i_{n+1}$.

We claim $a_{i_1} > a_{i_2} > \cdots > a_{i_{n+1}}$, a decreasing subsequence of length $n + 1$. If not, there is a pair with $a_{i_j} < a_{i_{j+1}}$. Then the increasing subsequence of length s ending at a_{i_j} can be extended by appending $a_{i_{j+1}}$, producing an increasing subsequence of length $s + 1$ ending at $a_{i_{j+1}}$. That contradicts $t(i_{j+1}) = s$. $\square$

Exercise. For any m and n, construct a sequence of mn distinct numbers with no increasing subsequence of length $m+1$ and no decreasing subsequence of length $n+1$. This shows the theorem is the strongest possible result.

1.5 Twenty Questions

One player thinks of an object and a second player asks yes/no questions until they can name it. How many distinct objects can be distinguished with twenty questions?

The strategy is *adaptive*: the k'th question may depend on the answers to the previous $k - 1$. But restrict the guesser to a deterministic strategy, so that a given sequence of answers always produces the same next question.

Even with adaptivity, the guesser must be able to tell any two objects apart. If two objects produced the same answers to every question, then — since the questions depend only on the previous answers — the guesser would reach the end of the twenty questions in an identical state for both and would have to guess, getting at least one of them wrong.

So each object determines a unique sequence of twenty yes/no answers. Take the objects as pigeons and the answer sequences as pigeonholes. There are 2^{20} sequences, so at most 2^{20} objects — a little over a million — can be distinguished. The bound is achieved: a strategy exists that separates 2^{20} objects in twenty questions.

> ***Note:*** *Allowing the guesser a probabilistic strategy does not change the largest number of objects that can be distinguished, though the argument is more involved and needs probability we have not yet developed.*

1.6 Twenty Questions with One Lie

Now suppose the answerer may lie once, a variation apparently first considered by Stanislaw Ulam. How many objects can the guesser distinguish?

Count the answer sequences compatible with a fixed object. The answerer may tell the truth throughout, or may lie on any one of the twenty questions — and once the lie is spent, every remaining answer is forced. That gives 21 distinct sequences per object. With t objects there are $21t$ pigeons and still 2^{20} pigeonholes.

If $21t > 2^{20}$, two pigeons land in the same hole. They cannot come from the same object, since the 21 sequences belonging to one object are distinct, so some answer sequence is compatible with two different objects and the guesser cannot separate them. The maximum is therefore

$$t = \left\lfloor \frac{2^{20}}{21} \right\rfloor = 49932.$$

> ***Note:*** *This is an upper bound, and whether a strategy attains it is a separate question. For some numbers of questions such a strategy exists, and for the rest one can still come reasonably close. The tools for settling this come later in the course, with error-correcting codes — a lie is exactly a transmission error, and the guesser's problem is to design questions robust to one of them.*

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